

INHA UNIVERSITY TASHKENT

FALL SEMESTER 2024

SOC 3010

OPERATING SYSTEM

CREDITS/HOURS PER WEEK : 3/3

COURSE TYPE : TECHNICAL CORE SEQUENCE

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SOC 3010

OPERATING SYSTEM

WEEK 1 LECTURE

INTRODUCTION TO OPERATING SYSTEM

Lecture Outline

- ★ **Introduction to Operating System**
 - **Basic Concepts**
 - **Functions**
 - **Types**
 - **Services**
 - **Properties**
 - **Process**



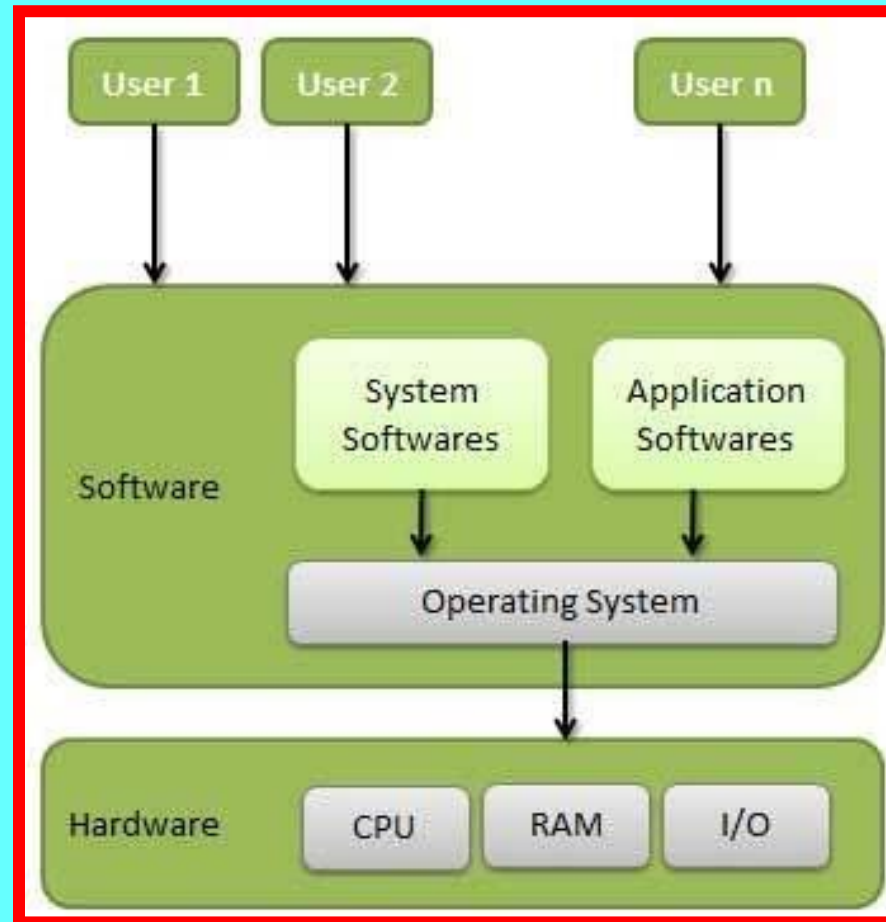
OPERATING SYSTEM – Basic Concepts

- ★ **An Operating System (OS) is an interface between a computer user and computer hardware.**
- ★ **An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.**
- ★ **Some popular Operating Systems include UNIX, Linux, Windows, OS X, VMS, OS/400, AIX, z/OS, etc.**

OPERATING SYSTEM – Basic Concepts

Definition

- * An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.



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OPERATING SYSTEM

WEEK 1 LECTURE 1

OPERATING SYSTEM FUNCTIONS

OPERATING SYSTEM -FUNCTIONS

❑ **Some important functions of an operating System are:**

- ★ **Memory Management**
- ★ **Processor Management**
- ★ **Device Management**
- ★ **File Management**
- ★ **Security**
- ★ **Control over system performance**
- ★ **Job accounting**
- ★ **Error detecting aids**
- ★ **Coordination between other software and users**

OPERATING SYSTEM -FUNCTIONS

❑ Memory Management

- ★ **Memory management refers to management of Primary Memory or Main Memory.**
- ★ Main memory is a large array of words or bytes where each word or byte has its own address.
- ★ Main memory provides a fast storage that can be accessed directly by the CPU. For a program to be executed, it must be in the main memory.
- ★ **An Operating System does the following activities for memory management –**
 - ❖ Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
 - ❖ In multiprogramming, the OS decides which process will get memory when and how much.
 - ❖ Allocates the memory when a process requests it to do so.
 - ❖ De-allocates the memory when a process no longer needs it or has been terminated.

OPERATING SYSTEM - FUNCTIONS

❑ Processor Management

- ★ In multiprogramming environment, the OS decides which process gets the processor when and for how much time.
- ★ This function is called **process scheduling**.
- ❑ An Operating System does the following activities for processor management –
 - ★ Keeps tracks of processor and status of process.
 - ★ The program responsible for this task is known as **traffic controller**.
 - ★ Allocates the processor (CPU) to a process.
 - ★ De-allocates processor when a process is no longer required.

OPERATING SYSTEM -FUNCTIONS

❑ Device Management

- ★ **An Operating System manages device communication via their respective drivers.**
- ★ **OS does the following activities for device management –**
 - ❖ **Keeps tracks of all devices.**
 - ❖ **Program responsible for this task is known as the I/O controller.**
 - ❖ **Decides which process gets the device when and for how much time.**
 - ❖ **Allocates the device in the efficient way.**
 - ❖ **De-allocates devices.**

OPERATING SYSTEM -FUNCTIONS

❑ File Management

- ★ A file system is normally organized into directories for easy navigation and usage.
- ★ These directories may contain files and other directories.
- ★ An Operating System does the following activities for file management –
 - ❖ Keeps track of file information, location, uses, status etc. The collective facilities are often known as **file system**.
 - ❖ Decides who gets the resources.
 - ❖ Allocates the resources.
 - ❖ De-allocates the resources.

OPERATING SYSTEM -FUNCTIONS

❑ Other Important Activities

- ❖ Some of the important activities that an Operating System performs :
- ★ **Security** – By means of password and similar other techniques, it prevents unauthorized access to programs and data.
- ★ **Control over System Performance** – Recording delays between request for a service and response from the system.
- ★ **Job accounting** – Keeping track of time and resources used by various jobs and users.
- ★ **Error detecting aids** – Production of dumps, traces, error messages, and other debugging and error detecting aids.
- ★ **Coordination between other Softwares and Users** – Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.

OPERATING SYSTEM -FUNCTIONS

❑ Questions

- * What is an OS?
- * What are the important functions performed by an Operating System?
- * Explain the following functions of an Operating System clearly listing the activities performed :
 - ❖ Memory Management
 - ❖ Processor Management
 - ❖ Device Management
 - ❖ File Management
 - ❖ Security
 - ❖ Control over System Performance
 - ❖ Job Accounting
 - ❖ Error Detecting Aids
 - ❖ Coordination between other Software and Users

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OPERATING SYSTEM

OPERATING SYSTEM TYPES

OPERATING SYSTEM TYPES

Operating systems are there from the very first computer generation and they keep evolving with time.

Some of the important types of operating systems which are most commonly used are:

- Batch operating system
- Time-sharing operating systems
- Distributed operating System
- Network operating System
- Real Time operating System

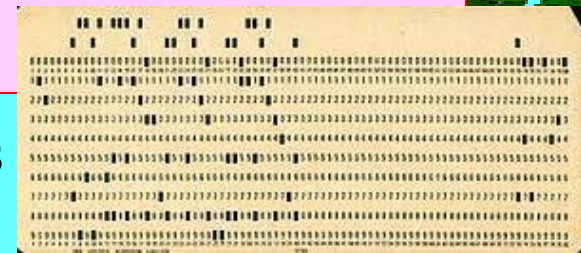
OPERATING SYSTEM TYPES

Punch Card in
Punch Card Machine



❑ Batch operating system

- ★ The users of a batch operating system do not interact with the computer directly.
- ★ Each user prepares his job on an off-line device like punch cards and submits it to the computer operator.
- ★ To speed up processing, jobs with similar needs are batched together and run as a group.
- ★ The programmers leave their programs with the operator and the operator then sorts the programs with similar requirements into batches.
- ★ The problems with Batch Systems are as follows –
 - Lack of interaction between the user and the job.
 - CPU is often idle, because the speed of the mechanical I/O devices is slower than the CPU.
 - Difficult to provide the desired priority.



OPERATING SYSTEM TYPES

❑ Time-sharing operating systems

- ✱ Time-sharing is a technique which enables many people, located at various terminals, to use a particular computer system at the same time.
- ✱ Time-sharing or multitasking is a logical extension of multiprogramming.
- ✱ Processor's time which is shared among multiple users simultaneously is termed as time-sharing.
- ✱ The main difference between Multiprogrammed Batch Systems and Time-Sharing Systems is that in case of Multiprogrammed batch systems, the objective is to maximize processor use, whereas in Time-Sharing Systems, the objective is to minimize response time.
- ✱ Multiple jobs are executed by the CPU by switching between them, but the switches occur so frequently.
- ✱ Thus, the user can receive an immediate response. For example, in a transaction processing, the processor executes each user program in a short burst or quantum of computation.
- ✱ That is, if n users are present, then each user can get a time quantum. When the user submits the command, the response time is in few seconds at most.

OPERATING SYSTEM TYPES

❑ Time-sharing operating systems

- ★ The operating system uses CPU scheduling and multiprogramming to provide each user with a small portion of a time.
- ★ Computer systems that were designed primarily as batch systems have been modified to time-sharing systems.
- ★ Advantages of Timesharing operating systems are as follows –
 - Provides the advantage of quick response.
 - Avoids duplication of software.
 - Reduces CPU idle time.
- ★ Disadvantages of Time-sharing operating systems are as follows –
 - Problem of reliability.
 - Question of security and integrity of user programs and data.
 - Problem of data communication.

OPERATING SYSTEM TYPES

❑ Distributed operating System

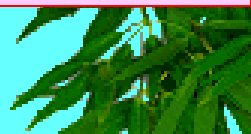
- * Distributed systems use multiple central processors to serve multiple real-time applications and multiple users.
- * Data processing jobs are distributed among the processors accordingly.
- * The processors communicate with one another through various communication lines (such as high-speed buses or telephone lines).
- * These are referred as loosely coupled systems or distributed systems.
- * Processors in a distributed system may vary in size and function.
- * These processors are referred as sites, nodes, computers, and so on.
- * The advantages of distributed systems are as follows –
 - With resource sharing facility, a user at one site may be able to use the resources available at another.
 - Speedup the exchange of data with one another via electronic mail.
 - If one site fails in a distributed system, the remaining sites can potentially continue operating.
 - Better service to the customers.
 - Reduction of the load on the host computer.
 - Reduction of delays in data processing.

OPERATING SYSTEM TYPES



❑ Network operating System

- ★ A Network Operating System runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions.
- ★ The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network, typically a local area network (LAN), a private network or to other networks.
- ★ Examples of network operating systems include Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, and BSD.
- ★ The advantages of network operating systems are as follows –
 - Centralized servers are highly stable.
 - Security is server managed.
 - Upgrades to new technologies and hardware can be easily integrated into the system.
 - Remote access to servers is possible from different locations and types of systems.
- ★ The disadvantages of network operating systems are as follows –
 - High cost of buying and running a server.
 - Dependency on a central location for most operations.
 - Regular maintenance and updates are required



OPERATING SYSTEM TYPES

❑ Real Time Operating System(RTOS)

- * A real-time system is defined as a data processing system in which the time interval required to process and respond to inputs is so small that it controls the environment.
- * The time taken by the system to respond to an input and display of required updated information is termed as the response time. So in this method, the response time is very less as compared to online processing.
- * Real-time systems are used when there are rigid time requirements on the operation of a processor or the flow of data and real-time systems can be used as a control device in a dedicated application.
- * A real-time operating system must have well-defined, fixed time constraints, otherwise the system will fail. For example, Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.
- * There are two types of real-time operating systems.
 - Hard real-time systems
 - Hard real-time systems guarantee that critical tasks complete on time. In hard real-time systems, secondary storage is limited or missing and the data is stored in ROM. In these systems, virtual memory is almost never found.
 - Soft real-time systems
 - Soft real-time systems are less restrictive. A critical real-time task gets priority over other tasks and retains the priority until it completes. Soft real-time systems have limited utility than hard real-time systems. For example, multimedia, virtual reality, Advanced Scientific Projects like undersea exploration and planetary rovers, etc.

OPERATING SYSTEM -TYPES

❑ Questions

- ★ List the types of operating Systems
- ★ Explain the following Operating System Types
 - Batch Operating System
 - Time-sharing Operating System
 - Distributed Operating System
 - Network Operating System
 - Real-Time Operating System (RTOS)

OPERATING SYSTEM -TYPES

❑ Questions

- ★ State the problems associated with Batch Systems
- ★ State the advantages and disadvantages of Time-Sharing Operating Systems
- ★ State the advantages of Distributed Systems.
- ★ State the advantages and disadvantages of Network Operating Systems.
- ★ What are the two types of Real-Time Operating Systems(RTOS)?
- ★ Distinguish between Hard Real-Time System and Soft Real-time System.

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OPERATING SYSTEM

OPERATING SYSTEM SERVICES

OPERATING SYSTEM SERVICES

- ★ An Operating System provides services to both the users and to the programs.
- ★ It provides programs an environment to execute.
- ★ It provides users the services to execute the programs in a convenient manner.
- ★ Following are a few common services provided by an operating system
 - Program execution
 - I/O operations
 - File System manipulation
 - Communication
 - Error Detection
 - Resource Allocation
 - Protection

OPERATING SYSTEM SERVICES

★ Program execution

- ★ Operating systems handle many kinds of activities from user programs to system programs like printer spooler, name servers, file server, etc.
- ★ Each of these activities is encapsulated as a process.
- ★ A process includes the complete execution context (code to execute, data to manipulate, registers, OS resources in use).
- ★ Following are the major activities of an operating system with respect to program management –
 - Loads a program into memory.
 - Executes the program.
 - Handles program's execution.
 - Provides a mechanism for process synchronization.
 - Provides a mechanism for process communication.
 - Provides a mechanism for deadlock handling.

OPERATING SYSTEM SERVICES

❑ I/O Operation

- ★ An I/O subsystem comprises of I/O devices and their corresponding driver software.
- ★ Drivers hide the peculiarities of specific hardware devices from the users.
- ★ An Operating System manages the communication between user and device drivers.
- ★ I/O operation means read or write operation with any file or any specific I/O device.
- ★ Operating system provides the access to the required I/O device when required.

OPERATING SYSTEM SERVICES

❑ File system manipulation

- * A file represents a collection of related information.
- * Computers can store files on the disk (secondary storage), for long-term storage purpose.
- * Examples of storage media include magnetic tape, magnetic disk and optical disk drives like CD, DVD.
- * Each of these media has its own properties like speed, capacity, data transfer rate and data access methods.
- * A file system is normally organized into directories for easy navigation and usage.
- * These directories may contain files and other directories.
- * Following are the major activities of an operating system with respect to file management –
 - Program needs to read a file or write a file.
 - The operating system gives the permission to the program for operation on file.
 - Permission varies from read-only, read-write, denied and so on.
 - Operating System provides an interface to the user to create/delete files.
 - Operating System provides an interface to the user to create/delete directories.
 - Operating System provides an interface to create the backup of file system.

OPERATING SYSTEM SERVICES

❑ Communication

- ★ In case of distributed systems which are a collection of processors that do not share memory, peripheral devices, or a clock, the operating system manages communications between all the processes.
- ★ Multiple processes communicate with one another through communication lines in the network.
- ★ The OS handles routing and connection strategies, and the problems of contention and security.
- ★ Following are the major activities of an operating system with respect to communication –
 - Two processes often require data to be transferred between them
 - Both the processes can be on one computer or on different computers, but are connected through a computer network.
 - Communication may be implemented by two methods, either by Shared Memory or by Message Passing.

OPERATING SYSTEM SERVICES

❑ Error handling

- ★ **Errors can occur anytime and anywhere.**
- ★ **An error may occur in CPU, in I/O devices or in the memory hardware.**
- ★ **Following are the major activities of an operating system with respect to error handling**
 - **The OS constantly checks for possible errors.**
 - **The OS takes an appropriate action to ensure correct and consistent computing.**

OPERATING SYSTEM SERVICES

❑ Resource Management

- ★ In case of multi-user or multi-tasking environment, resources such as main memory, CPU cycles and files storage are to be allocated to each user or job.
- ★ Following are the major activities of an operating system with respect to resource management –
 - The OS manages all kinds of resources using schedulers.
 - CPU scheduling algorithms are used for better utilization of CPU.

OPERATING SYSTEM SERVICES

❑ Protection

- ★ Considering a computer system having multiple users and concurrent execution of multiple processes, the various processes must be protected from each other's activities.
- ★ Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by a computer system.
- ★ Following are the major activities of an operating system with respect to protection –
 - The OS ensures that all access to system resources is controlled.
 - The OS ensures that external I/O devices are protected from invalid access attempts.
 - The OS provides authentication features for each user by means of passwords.

OPERATING SYSTEM -SERVICES

□ Questions

★ Describe the Operating System Services

★ Explain the following Operating System Services

- Program execution
- I/O operations
- File System manipulation
- Communication
- Error Detection
- Resource Allocation
- Protection

OPERATING SYSTEM -SERVICES

❑ Questions

- ★ List the major activities performed by the operating system with respect to program management.
- ★ List the major activities performed by the operating system with respect to file management.
- ★ List the major activities performed by the operating system with respect to communication.
- ★ List the major activities performed by the operating system with respect to error handling.
- ★ List the major activities performed by the operating system with respect to resource management.
- ★ List the major activities performed by the operating system with respect to protection.

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OPERATING SYSTEM

OPERATING SYSTEM PROPERTIES

OPERATING SYSTEM PROPERTIES

❑ PROPERTIES AN OPERATING SYSTEM SHOULD POSSESS :

- Batch Processing
- Multi-Tasking
- Muti-Progamming
- Interactivity
- Real-Time Processing
- Distributed Environment
- Spooling

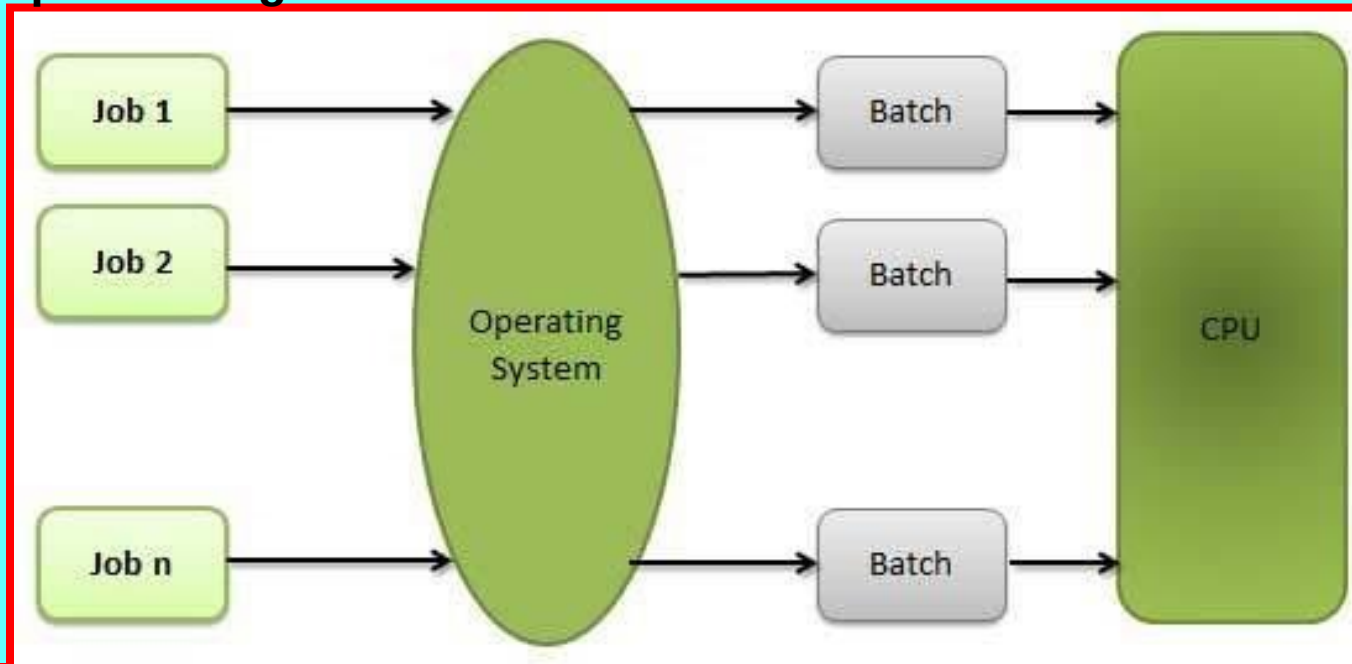
OPERATING SYSTEM PROPERTIES

❑ Batch processing

- ★ Batch processing is a technique in which an Operating System collects the programs and data together in a batch before processing starts.
- ★ An operating system does the following activities related to batch processing –
 - The OS defines a job which has predefined sequence of commands, programs and data as a single unit.
 - The OS keeps a number of jobs in memory and executes them without any manual intervention.
 - Jobs are processed in the order of submission, i.e., first come first served fashion.
 - When a job completes its execution, its memory is released and the output for the job gets copied into an output spool for later printing or processing.

OPERATING SYSTEM PROPERTIES

❑ Batch processing



Advantages

- * Batch processing takes much of the work of the operator to the computer.
- * Increased performance as a new job get started as soon as the previous job is finished, without any manual intervention.

Disadvantages

- * Difficult to debug program.
- * A job could enter an infinite loop.
- * Due to lack of protection scheme, one batch job can affect pending jobs.

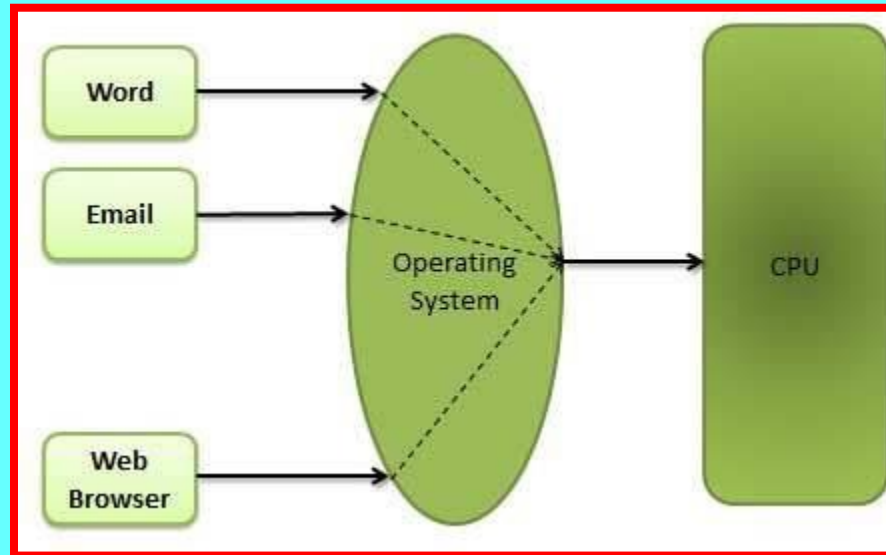
OPERATING SYSTEM PROPERTIES

❑ Multitasking

- * Multitasking is when multiple jobs are executed by the CPU simultaneously by switching between them.
- * Switches occur so frequently that the users may interact with each program while it is running.
- * An OS does the following activities related to multitasking –
 - The user gives instructions to the operating system or to a program directly, and receives an immediate response.
 - The OS handles multitasking in the way that it can handle multiple operations/executes multiple programs at a time.
 - Multitasking Operating Systems are also known as Time-sharing systems.
 - These Operating Systems were developed to provide interactive use of a computer system at a reasonable cost.
 - A time-shared operating system uses the concept of CPU scheduling and multiprogramming to provide each user with a small portion of a time-shared CPU.
 - Each user has at least one separate program in memory.

OPERATING SYSTEM PROPERTIES

❑ Multitasking



- * A program that is loaded into memory and is executing is commonly referred to as a process.
- * When a process executes, it typically executes for only a very short time before it either finishes or needs to perform I/O.
- * Since interactive I/O typically runs at slower speeds, it may take a long time to complete.
- * During this time, a CPU can be utilized by another process.

OPERATING SYSTEM PROPERTIES

❑ Multitasking

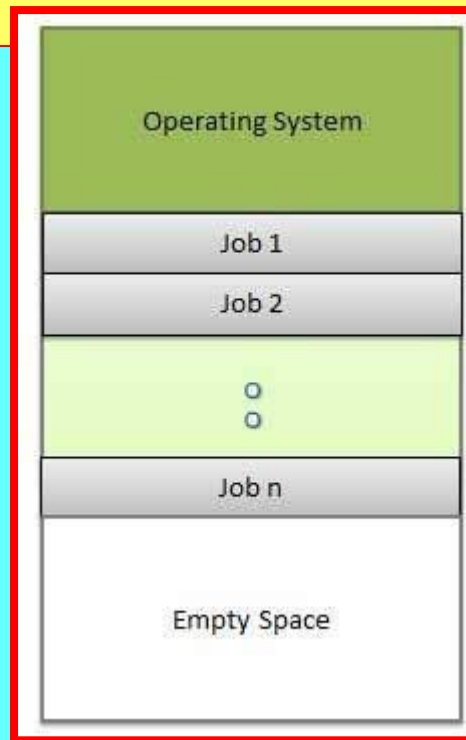
- ★ The operating system allows the users to share the computer simultaneously.
- ★ Since each action or command in a time-shared system tends to be short, only a little CPU time is needed for each user.
- ★ As the system switches CPU rapidly from one user/program to the next, each user is given the impression that he/she has his/her own CPU, whereas actually one CPU is being shared among many users.

OPERATING SYSTEM PROPERTIES

❑ Multiprogramming

- ★ Sharing the processor, when two or more programs reside in memory at the same time, is referred as multiprogramming.
- ★ Multiprogramming assumes a single shared processor.
- ★ Multiprogramming increases CPU utilization by organizing jobs so that the CPU always has one to execute.

Memory layout for a multiprogramming system



OPERATING SYSTEM PROPERTIES

❑ Multiprogramming

- * An OS does the following activities related to multiprogramming.
 - The operating system keeps several jobs in memory at a time.
 - This set of jobs is a subset of the jobs kept in the job pool.
 - The operating system picks and begins to execute one of the jobs in the memory.
 - Multiprogramming operating systems monitor the state of all active programs and system resources using memory management programs to ensure that the CPU is never idle, unless there are no jobs to process.
- * **Advantages**
 - High and efficient CPU utilization.
 - User feels that many programs are allotted to CPU almost simultaneously.
- * **Disadvantages**
 - CPU scheduling is required.
 - To accommodate many jobs in memory, memory management is required.

OPERATING SYSTEM PROPERTIES

❑ Interactivity

- ★ **Interactivity refers to the ability of users to interact with a computer system.**
- ★ **An Operating system does the following activities related to interactivity –**
 - Provides the user an interface to interact with the system.
 - Manages input devices to take inputs from the user. For example, keyboard.
 - Manages output devices to show outputs to the user. For example, Monitor.
 - The response time of the OS needs to be short, since the user submits and waits for the result.

OPERATING SYSTEM PROPERTIES

❑ Real Time System

- ★ Real-time systems are usually dedicated, embedded systems.
- ★ An operating system does the following activities related to real-time system :
 - In such systems, Operating Systems typically read from and react to sensor data.
 - The Operating system must guarantee response to events within fixed periods of time to ensure correct performance.

OPERATING SYSTEM PROPERTIES

❑ Distributed Environment

- ★ A distributed environment refers to multiple independent CPUs or processors in a computer system.
- ★ An operating system does the following activities related to distributed environment –
 - The OS distributes computation logics among several physical processors.
 - The processors do not share memory or a clock. Instead, each processor has its own local memory.
 - The OS manages the communications between the processors.
 - They communicate with each other through various communication lines.

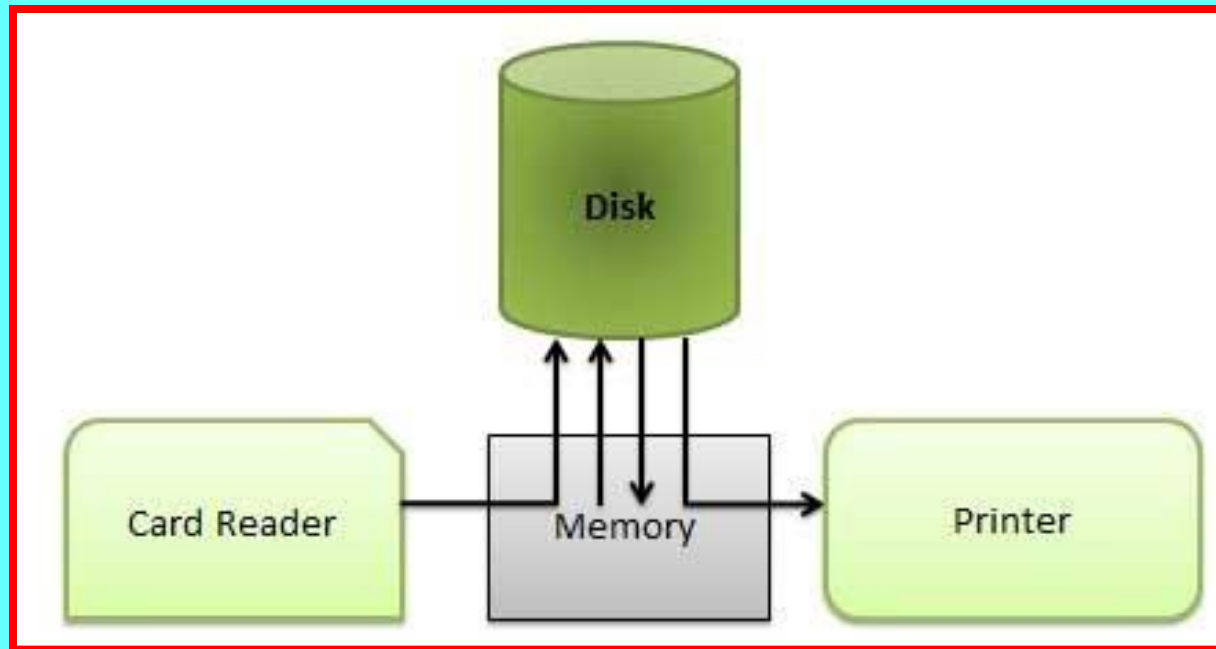
OPERATING SYSTEM PROPERTIES

❑ Spooling

- ✱ SPOOL is an acronym for Simultaneous Peripheral Operations On Line.
- ✱ Spooling refers to putting data of various I/O jobs in a buffer.
- ✱ This buffer is a special area in memory or hard disk which is accessible to I/O devices.
- ✱ An operating system does the following activities related to Spooling :
 - Handles I/O device data spooling as devices have different data access rates.
 - Maintains the spooling buffer which provides a waiting station where data can rest while the slower device catches up.
 - Maintains parallel computation because of spooling process a computer can perform I/O in parallel fashion.
 - It becomes possible to have the computer read data from a tape, write data to disk and to write out to a tape printer while it is doing its computing task.

OPERATING SYSTEM PROPERTIES

❑ Spooling



★ Advantages

- The spooling operation uses a disk as a very large buffer.
- Spooling is capable of overlapping I/O operation for one job with processor operations for another job.

OPERATING SYSTEM -PROPERTIES

❑ Questions

- ★ List the properties an Operating System should possess.
- ★ Explain the following Operating System properties
 - Batch Processing
 - Multi-tasking
 - Multi-programming
 - Real-Time System
 - Distributed Environment
 - Spooling
- ★ List the activities performed by the operating system related to Batch Processing.
- ★ State the advantages and disadvantages of Batch Processing.

OPERATING SYSTEM -PROPERTIES

❑ Questions

- ★ List the activities performed by the operating system related to Multi-tasking.
- ★ State the advantages and disadvantages of Multiprogramming.
- ★ List the activities performed by the operating system related to Interactivity.
- ★ List the activities performed by the operating system related to Real-time System.
- ★ List the activities performed by the operating system related to Distributed Environment.
- ★ List the activities performed by the operating system related to Spooling.
- ★ What is the full form of SPOOL?
- ★ What is the need for Spooling?
- ★ State the advantages of Spooling.

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OPERATING SYSTEM

PROCESS

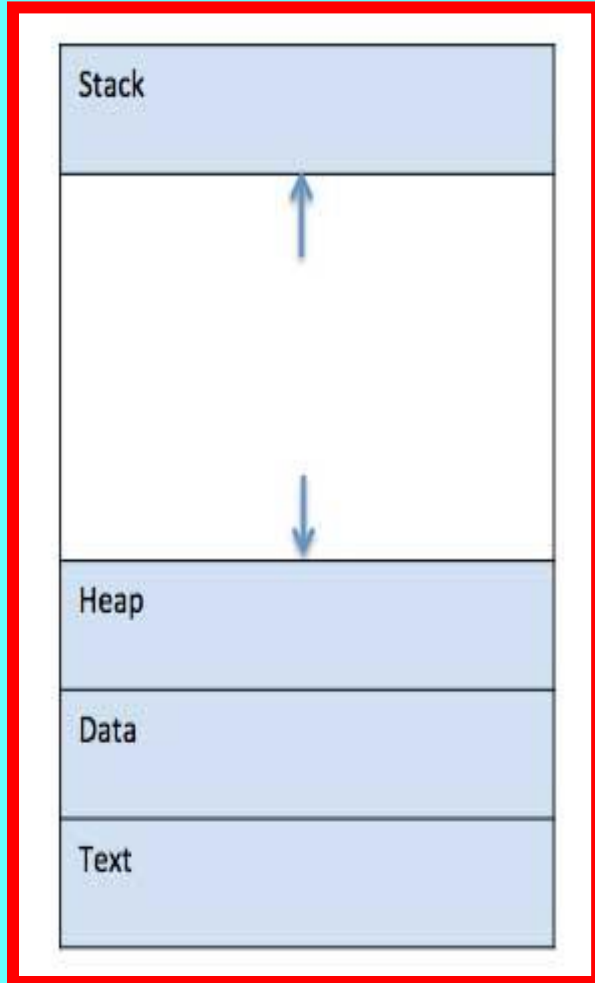
OPERATING SYSTEM - PROCESS

□ Process

- ★ A process is basically a program in execution.
- ★ The execution of a process must progress in a sequential fashion.
- ★ A process is defined as an entity which represents the basic unit of work to be implemented in the system.
- ★ To put it in simple terms, we write our computer programs in a text file and when we execute this program, it becomes a process which performs all the tasks mentioned in the program.
- ★ When a program is loaded into the memory and it becomes a process, it can be divided into four sections — **stack, heap, text and data.**

OPERATING SYSTEM - PROCESS

A simplified layout of a process inside main memory



Stack

The process Stack contains the temporary data such as method/function parameters, return address and local variables.

Heap

This is dynamically allocated memory to a process during its run time

Text

This includes the current activity represented by the value of Program Counter and the contents of the processor's registers.

Data

This section contains the global and static variables

OPERATING SYSTEM - PROCESS

□ Program

- ★ A program is a piece of code which may be a single line or millions of lines.
- ★ A computer program is usually written by a computer programmer in a programming language.

- ★ For example, here is a simple program written in C programming language –

```
#include <stdio.h>
int main()
{
    printf("Hello, World! /n");
    return 0;
}
```

- ★ A computer program is a collection of instructions that performs a specific task when executed by a computer.
- ★ When we compare a program with a process, we can conclude that a **process is a dynamic instance of a computer program.**
- ★ A part of a computer program that performs a well-defined task is known as an **algorithm**. A collection of computer programs, libraries and related data are referred to as a **software**.

OPERATING SYSTEM - PROCESS

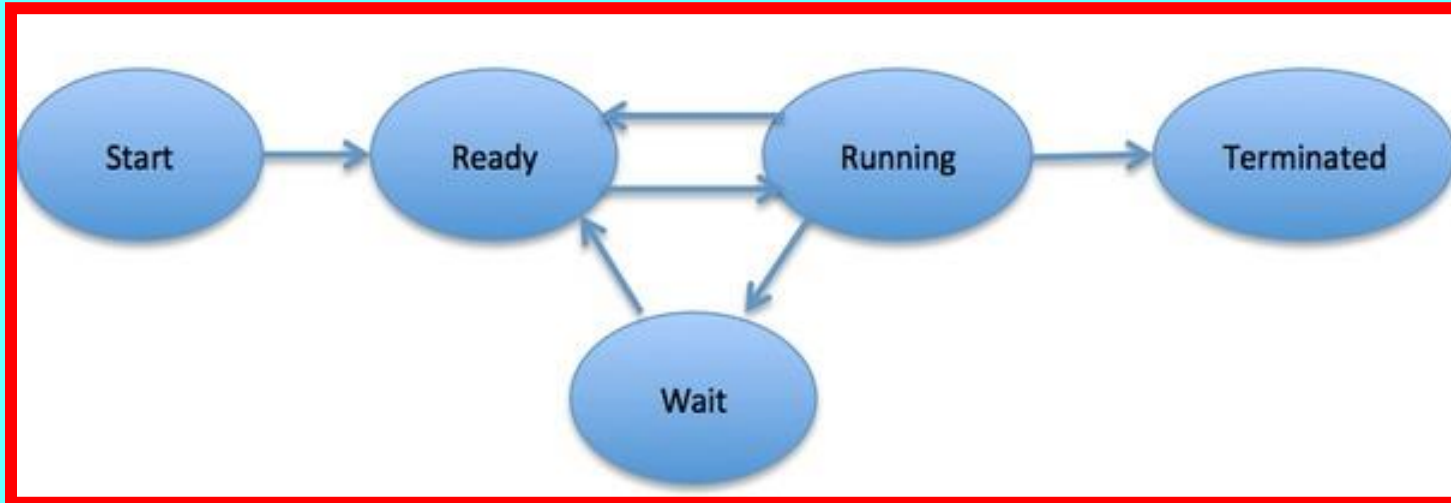
❑ Process Life Cycle

- ★ When a process executes, it passes through different states.
- ★ These stages may differ in different operating systems, and the names of these states are also not standardized.
- ★ In general, a process can have one of the following five states at a time :

Start, Ready, Running, Waiting, Terminated or Exit

OPERATING SYSTEM - PROCESS

□ Process Life Cycle



Start

This is the initial state when a process is first started/created

Running

Once the process has been assigned to a processor by the OS scheduler, the process state is set to running and the processor executes its instructions.

Waiting

Process moves into the waiting state if it needs to wait for a resource, such as waiting for user input, or waiting for a file or I/O to become available.

Ready

The process is waiting to be assigned to a processor. Ready processes are waiting to have the processor allocated to them by the operating system so that they can run. Process may come into this state after Start state or from running state interrupted by the scheduler to assign CPU to some other process

Terminated or Exit

Once the process finishes its execution, or it is terminated by the operating system, it is moved to the terminated state where it waits to be removed from main memory.

OPERATING SYSTEM - PROCESS

❑ Process Control Block (PCB)

- ★ A Process Control Block (also termed as Process descriptor) is a data structure maintained by the Operating System for every process.
- ★ The PCB is identified by an integer process ID (PID).
- ★ A PCB keeps all the information needed to keep track of a process.

OPERATING SYSTEM - PROCESS

S.N	Process Control Block (PCB) Information & Description
1	Process State The current state of the process i.e., whether it is ready, running, waiting, or whatever.
2	Process privileges This is required to allow/disallow access to system resources.
3	Process ID Unique identification for each of the process in the operating system.
4	Pointer A pointer to parent process.
5	Program Counter Program Counter is a pointer to the address of the next instruction to be executed for this process.
6	CPU registers Various CPU registers where process need to be stored for execution for running state.
7	CPU Scheduling Information Process priority and other scheduling information which is required to schedule the process.
8	Memory management information This includes the information of page table, memory limits, Segment table depending on memory used by the operating system.
9	Accounting information This includes the amount of CPU used for process execution, time limits, execution ID etc.
10	IO status information This includes a list of I/O devices allocated to the process.

OPERATING SYSTEM - PROCESS

❑ Process Control Block (PCB)

The architecture of a PCB is completely dependent on Operating System and may contain different information in different operating systems. Here is a simplified diagram of a PCB



The PCB is maintained for a process throughout its lifetime, and is deleted once the process terminates.

OPERATING SYSTEM - PROCESS

❑ Questions

- ★ What is a process?
- ★ Process is **(SELECT ONE)**
 - a) program in High level language kept on disk
 - b) contents of main memory
 - c) a program in execution
 - d) a job in secondary memory
- ★ Distinguish between Program & Process.
- ★ Describe the Process Life Cycle using a State diagram
- ★ What is Process Control Block? What information does it contain?

SOC 3010

OPERATING SYSTEM

END OF
WEEK1 LECTURE

INTRODUCTION TO
OPERATING SYSTEM

WISH YOU ALL THE BEST

INHA UNIVERSITY TASHKENT

FALL SEMESTER 2024

SOC 3010

OPERATING SYSTEM

CREDITS/HOURS PER WEEK : 3/3

COURSE TYPE : TECHNICAL CORE SEQUENCE

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